Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY



A CASE STUDY REPORT
ON

"Vision-Based Printing Defect Detection and Sorting System"

PREPARED
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DEPARTMENT OF MECHATRONICS ENGINEERING

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Department of Mechatronics Engineering



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This report is submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Mechatronics Engineering at Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh.

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Rajshahi University of Engineering & Technology Rajshahi

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Certificate

The present project report is to certify that Vision-Based Printing Defect Detection and Sorting System is conducted by Sumaiya Rahman Oishy-2008037, Madhurima Das-2008038, Nawshin Nawar-2008039, Tanjim Samad Swapnil-2008041, Al Nimra Kabir-2008042 under my supervision within the Department of Mechatronics Engineering at Rajshahi University of Engineering and Technology.

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Abstract

The project is called Vision-Based Printing Defect Detection and Sorting System and introduces a more realistic solution to apply the process of quality inspection to automation in the manufacturing industry, namely, the detection and sorting of defects in printed materials. This is a system that combines machine vision and image processing with automated sorting systems to enhance accuracy, speed and efficiency in industrial quality control. The project will be focused on using high resolution cameras and image analysis algorithms to identify defect in product printing as misalignment, smudges, fading, and pattern variances on the printed surfaces. Defects are detected and the system activates actuators to separate defect and acceptable items in real-time. As it is implemented, it emphasizes the increasing importance of vision-based systems in contemporary manufacturing, specifically in lessening human error, enhanced throughput, and stable quality of a product. Moreover, the project indicates the implementation of the concepts of mechatronics that includes hardware and software integration, control systems, and sensing technologies to address practical industrial issues. It also correlates with the visions of Industry 4.0, which advances automation, smart surveillance, and evidence-based decision-making. The work can be used to further develop intelligent inspection systems and, in general, shows potential of mechatronics engineering in promoting industrial automation.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CPU	Central Processing Unit
CNN	Convolutional Neural Network
DL	Deep Learning
FPS	Frames Per Second
GPU	Graphics Processing Unit
IoT	Internet of Things
ML	Machine Learning
OCR	Optical Character Recognition
PR	Precision-Recall
QA	Quality Assurance
ROI	Region of Interest
RSIs	Repetitive Strain Injuries
RUET	Rajshahi University of Engineering & Technology
SDG	Sustainable Development Goals
YOLO	You Only Look Once
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

In the age of automation and smart systems, product quality and consistency had turned into an important feature of the production process, especially in such a sphere as pharmaceuticals where accuracy was the primary consideration. The project was called Vision-Based Printing Defect Detection and Sorting System which aimed at designing an intelligent machine learning based system to detect and sort defective packets in real time, specifically, misprinted packets, of product packets. To do this the custom dataset and machine learning model was trained on the images of medicine packets and allowed the system to learn and identify correct and incorrect print patterns. The implementation was conducted with the help of Python as the initial object detection framework and OpenCV as the task image processing framework. This dataset is composed of JPEG images of medicine packets designs, and was hand-curated and annotated to suit the model. In contrast to the generic defect detection systems, this solution was customized to high-accuracy, real world deployment where speed and reliability mattered and that only high-quality and correctly labeled packets would reach the end consumer with the high importance of packaging impression.

1.1 Background Studies

Historically the quality control of the packaging had strongly depended on manual inspection which was carried out by trained operators. Although human inspection was thought to be functional in small scale production, it had become extremely inefficient, inconsistent, and error prone in high-speed, high-volume production settings. Factors like fatigue of operators, subjectivity and differences in lighting conditions had made manual inspection an ineffective way of maintaining a uniform quality standard. This meant that there was now a growing need to have automated and intelligent inspection systems that could be used to detect the existence of tiny defects which were not visible by the human eye in visual patterns with a high degree of precision. Recent developments in the computer vision and machine learning had provided opportunities in the development of automated inspection systems that could be used to detect the existence of minor defects that could not be spotted by the human eye in the visual pattern. Computer vision methods had already been used to remove meaningful visual representations of images, and machine learning algorithms specifically convolutional neural networks (CNNs) had been applied to perform tasks of pattern recognition and classification on the scale of human perception. These technologies were becoming increasingly used in applications like optical character recognition (OCR), as well as defect detection in manufacturing lines. Object detection models like YOLO (You Only Look Once), Faster R-CNN and SSD (Single Shot Multi-Box Detector), among others. using image processing software, such as OpenCV had transformed visual data processing and interpretation. Trained on domain specific datasets, the models had already attained high degrees of accuracy on detecting anomalies, even when the

lighting and perspective varied.

In this respect, the project was conducted with the aim of filling the gap existing between theoretical research and actual implementation by establishing a tailor-made machine learning based system of identifying and classifying printing flaws on pharmaceutical packaging. A trained model was trained on a hand-curated collection of medicine packet images to detect common print related defects in real time and signal relevant sorting mechanisms. The aim of the approach was to not only improve the efficiency of the production lines but also to reduce the chances of defective products reaching the market hence improving their compliance with the industry standards and consumer safety.

The background analysis provided above had actually underpinned the creation and deployment of the "Vision Based Printing Defect Detection and Sorting System" by pointing to the weakness of the traditional quality control paradigm and the opportunities afforded by the recent computer vision and machine learning tools.

On the one hand, the paper had stressed the vital role of packaging the accuracy of pharmaceutical products, which coincided with the objective of the project to detect and remove misprinted or defective medicine packets. In this field, the slightest mistake in labeling or the quality of print might have caused major health and safety concerns. The system has dealt with this concern by providing an automated and data driven system that could identify such errors with high accuracy.

Secondly, the history had highlighted the ineffectiveness of manual inspection especially in the production chains that were fast moving. This is a factor that was of particular importance to the project that had sought to substitute manual processes that were slow and subject to errors with an automated and real time inspection system. Machine learning integration had guaranteed the quality control was consistent and scalable thus defeating subjectivity that accompanies human inspectors and the fatigue they experienced.

Moreover, the technical approach was preconditioned by the discussion of the methods of computer vision and machine learning. The project was constructed on established techniques of visual defects detection by using object detection frameworks and algorithms like PyTorch and OpenCV.

Finally, the background had brought out the increasing popularity of intelligent inspection systems use in the manufacturing industry, thus supporting the fact that the project was technically sound, timely, and befitting the current industry demands. This confirmed the choice of using a domain specific, customized model instead of using generic defect detection tools, which would guarantee the greater accuracy and enhanced flexibility in practical application to the real-world deployment.

1.2 Motivation

This project had been inspired by the internship experience at Walton, where it had been observed that packaging defect detection and sorting were carried out manually by workers. The process had been found to be time consuming, labor intensive, and prone to human manipulation, which had led to rise operational costs and ease efficiency. In industries such as pharmaceuticals, such errors in print alignment or labeling had posed serious risks, including regulatory violations and loss of consumer trust. To address these challenges, a vision based, machine learning solution had been proposed to automate the detection and sorting process. It had been expected that this system would significantly reduce labor costs, improve accuracy, and ensure consistent product quality. Moreover, the model had been designed with scalability in mind, making it applicable across various manufacturing sectors. As part of the broader movement toward Industry 4.0, such intelligent systems had been nominated as essential for enhancing industrial automation, secure quality control, and reducing production overhead in the long term.

1.3 Objectives

The objective of this project is to develop a machine learning based computer vision. That can accurately estimate printing defect like misalignments, miss prints, incorrect design on medicine packets.

- To develop a machine learning-based system capable of accurately detecting printing defects such as misalignments, faded text, or incorrect designs on medicine packets using computer vision techniques.
- To automate the sorting process of correctly printed and defective packets in real-time, reducing reliance on manual labor and minimizing human error on the production line.
- To design and demonstrate a prototype system of vision-based printing defect detection and sorting system.

1.4 Work Plan

The Gantt chart of the detailed work plan of the project is attached here in Figure 1.1. We got 12 weeks to complete our work, so we split our work so that we can complete it within time.

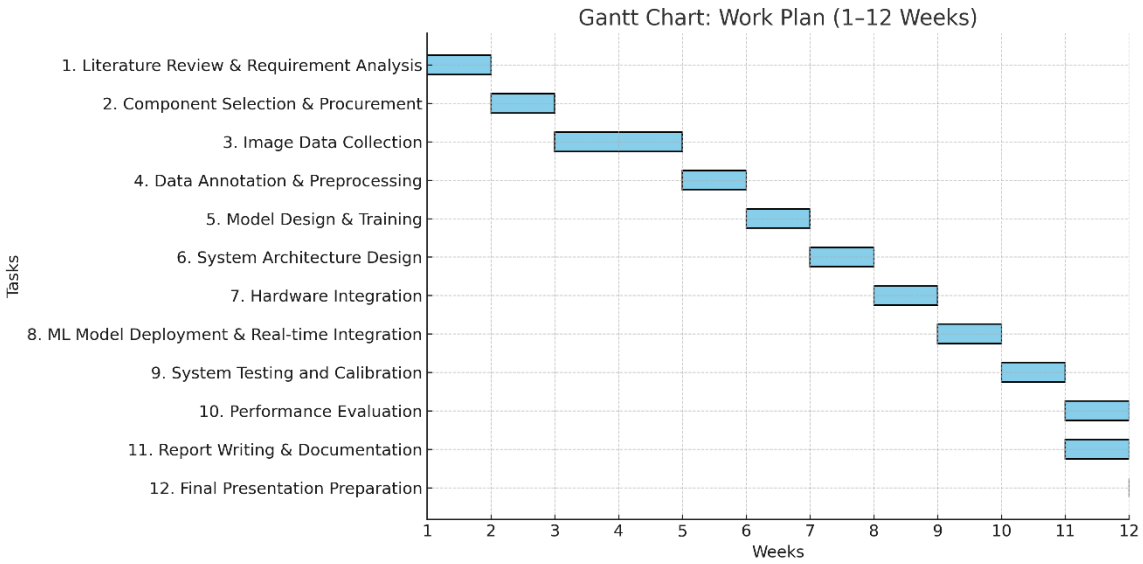


Fig 1.1: Gantt Chart of Project Work Plan.

1.5 Summary

This chapter presents a machine learning based product the Vision Based Printing Defect Detection and Sorting System, a machine learning based product that functions with the aim of automating the process of detecting and sorting defects in a package, especially pharmaceutical package. The system applies computer vision and machine learning algorithms, including CNNs and object detection models such as YOLO, Faster R-CNN and SSD to detect defects such as not perfectly aligned, faded prints, and incorrect design in real-time, producing high quality products. Driven by inefficiencies witnessed during an internship at Walton in which manual inspection had been time consuming and error prone, the system is set to cut labor costs, improve accuracy, and efficiency in production. The project is written in Python, PyTorch, and OpenCV, and it fulfills a dire need to handle the exact packaging in medications, errors in which may have critical health effects. The goals of the system: automation of the defect detection and sorting process, enhancement of production lines reliability, and proving the prospects of intelligent inspection systems in industries.

Chapter 2

Literature Review & Methodology

2.1 Introduction

Quality assurance (QA) in packaging systems is vital for ensuring product integrity, customer satisfaction, and regulatory compliance. Traditional inspection methods are manual, time-consuming, and prone to inconsistency. To address these challenges, recent studies have introduced machine learning (ML) and machine vision systems for automating visual inspections. This chapter reviews existing literature on automated QA using ML, highlights current research gaps, and explains the methodology adopted for this project which aims to develop an ML based real-time quality assurance system for packaging using image processing and defect detection.

2.2 Literature Review

2.2.1 Machine Vision Based Quality Inspection of Pharmaceutical Bottle Packaging

Bhoomika Shetty et al. applied machine vision and AI to detect defects in pharmaceutical bottle packaging. The method supported Pharma 4.0 by enabling real-time visual inspection. While the system was effective, it had limitations due to the use of black-box AI models, limited explainable AI integration, and difficulty in generalizing across variable datasets. [1]

2.2.2 A Systematic Literature Review on Artificial Intelligence and Explainable Artificial Intelligence for Visual Quality Assurance in Manufacturing

Rudolf Hoffmann and Christoph Reich conducted a review of AI and XAI in manufacturing visual inspection. Most work focused on defect classification without tackling interpretability or root cause analysis. The review emphasized a lack of transparency in most models. [2]

2.2.3 Visual Inspection in Additive Manufacturing Using GradCAM for Model Explainability

Alexandre M. A. Maciel used pretrained CNNs (like VGG16) with GradCAM to explain defect detection decisions in additive manufacturing. The method improved local interpretability by highlighting important image regions. However, it was limited to CNNs, lacked global model insight, and required access to internal model layers. [3]

2.2.4 CNN-Based Anomaly Detection and Root Cause Analysis for Predictive Maintenance

Rao X. and Zhang F. used CNNs combined with SHAP, GradCAM, and LIME to detect anomalies and identify root causes in predictive maintenance of mechanical systems. Their model provided interpretable insights but had high computational costs,

inconsistent XAI outputs, and required expert interpretation for explanation validation. [4].

2.2.5 Tree-Based Models for Root Cause Analysis in Additive Manufacturing

Herrmann C., Schmidt C., and Kurle used interpretable models like CART and XGBoost to analyze root causes of defects in 3D-printed components. The models enabled clear decision paths, aiding understanding. Still, they were less effective on raw image data and were sensitive to imbalanced datasets compared to deep learning models. [5]

2.2.6 Machine Vision for Detecting Defects in Liquid Bottles: An Industrial Application for Food and Packaging Sector

Omid Farhangi et al. implemented a real-time machine vision system using CMOS cameras and template-matching for liquid bottle inspection. It accurately checked for defects like cap misalignment and label errors in transparent bottles. However, it lacked explainable AI, was sensitive to environmental conditions, and was not applicable to opaque bottles. [6]

2.2.7 State of the Art in Defect Detection Based on Machine Vision

Li T. S. and Huang C. L. provided a broad review of machine vision applications for defect detection in electronics and ceramics. Their review emphasized the speed and non-destructive nature of vision systems. Challenges included processing complex patterns, high computational load, ethical concerns, and lack of standardized inspection protocols. [7]

2.2.8 An Efficient Fault Detection Method for Induction Motors Using Thermal Imaging and Machine Vision

Muhammad Rameez Javed et al. proposed a fault detection method for induction motors using thermal imaging and SVM classification. Their method enabled real- time, non-invasive detection of motor conditions like overload and faults. However, it required high-quality infrared cameras, had computational demands, and was limited in scope to induction motors. [8]

2.2.9 Fault Detection and Diagnosis in Industry 4.0: A Review on Challenges and Opportunities

Denis Leite et al. reviewed real-time fault detection and diagnosis (RT-FDD) using ML in Industry 4.0 systems. The review identified its value in predictive maintenance and industrial monitoring. Limitations included a shallow discussion of challenges, lack of future research directions, and missing comparisons within categories of challenges.

2.2.3 Literature Review Table

Paper Title	Year Published	Publisher	Authors	Applications and Limitations
1.Machine Vision Based Quality Inspection of Pharmaceutical Bottle Packaging	2024	*Research Journal of Pharmacy and Technology (RJPT)*	Bhoomika Shetty, Madhushree S M, Muddukrishna B.S, Ravindra Shenoy, Kirankumar H, Mahendra Joshi, Girish Pai K	**Applications:** Machine vision and AI used for defect detection in pharmaceutical bottle packaging. Focus on Pharma 4.0 integration. **Limitations:** Black-box AI models may lack transparency; limited use of explainable AI; model generalization can be difficult due to dataset variability.
2.A Systematic Literature Review on Artificial Intelligence and Explainable Artificial Intelligence for Visual Quality	2023	*Electronics (MDPI)*	Rudolf Hoffmann, Christoph	**Applications:** Review of AI and XAI methods for visual inspection in manufacturing sectors (e.g., electronics, packaging, food, etc.). Identifies use cases for quality control, predictive maintenance, and process optimization. **Limitations:** Only 8% of reviewed studies applied XAI; most studies focus

Assurance in Manufacturing			Reich	narrowly on classification without addressing root cause analysis or interpretability challenges.
3.Visual Inspection in Additive Manufacturing Using GradCAM for Model Explainability	2023		Alexandre M. A. Maciel	**Applications:** Used pretrained CNNs (e.g., VGG16) and GradCAM to visually explain classification decisions in powder bed additive manufacturing defects. **Limitations:** GradCAM is limited to CNNs; does not offer global model explainability; requires access to model internals.
4.CNN-Based Anomaly Detection and Root Cause Analysis for Predictive Maintenance	2023	MDPI	Rao, X., Zhang, F	**Applications:** CNN with SHAP, GradCAM, LIME for interpreting predictive maintenance decisions in turbofan engines and other mechanical components. **Limitations:** High computational cost; XAI methods sometimes inconsistent; requires expert knowledge to validate explanations.

5.Tree-Based Models for Root Cause Analysis in Additive Manufacturing	2022-2023	MDPI	Herrmann, C., Schmidt, C., Kurle	**Applications:** Used interpretable tree-based models (CART, XGBoost) to predict defect sources in 3D-printed parts. **Limitations:** Tree models may underperform vs. CNNs on raw image data; sensitive to data imbalance.
6.Machine Vision Based Quality Inspection of Pharmaceutical Bottle Packaging	2024	*Research Journal of Pharmacy and Technology (RJPT)*	Bhoomika Shetty, Madhushree S M, Muddukrishna B.S, Ravindra Shenoy, Kirankumar H, Mahendra Joshi, Girish Pai K	**Applications:** Machine vision and AI used for defect detection in pharmaceutical bottle packaging. Focus on Pharma 4.0 integration. **Limitations:** Black-box AI models may lack transparency; limited use of explainable AI; model generalization can be difficult due to dataset variability.
7.Machine Vision for Detecting Defects in Liquid Bottles: An Industrial Application for Food and Packaging Sector	2024	*Universal Wiser Publisher*	Omid Farhangi , Ehsan Sheidaee , Asma kisalaei	**Applications:** Real-time vision inspection using CMOS imaging,template-based algorithms, and distance/pattern matching for liquid, cap, and label checks **Limitations:** Applicability limited to transparent bottles; no XAI; environmental robustness and actuation integration not addressed

8. State of the Art in Defect Detection Based on Machine Vision	2022	*Springer*	Li, T. S., & Huang, C. L.	**Applications**: Comprehensive review of machine vision for defect detection in industries like electronics and ceramics. Supports high-speed, non-destructive testing. **Limitations**: Challenges with complex patterns; high computational costs; ethical concerns and need for standardized protocols.
9.An Efficient Fault Detection Method for Induction Motors Using Thermal Imaging and Machine Vision	2022	*MDPI*	Muhammad Rameez Javed, Zain Shabbir, Furqan Asghar, Waseem Amjad	**Applications**: - Detects induction motor faults using thermal imaging and machine vision. - Classifies motor states (normal, overloaded, fault) with IRT and SVM. - Enables non-invasive, real-time industrial maintenance. **Limitations**: - Limited to induction motors. - Requires high-quality infrared cameras. - Computational complexity may hinder real-time use. Sensitive to environmental conditions

10. Fault Detection and Diagnosis in Industry 4.0: A Review on Challenges and Opportunities	2025	*MDPI*	Denis Leite, Emmanuel Andrade, Diego Rativa, Alexandre M. A. Maciel	**Applications**: - Identifies innovative fault detection methods across industries. - Supports predictive maintenance and real-time monitoring. **Limitations**: - Limited in-depth discussion of challenges and solutions. Lacks specific future research directions.

2.3 Research Gap

Despite substantial progress in the development of machine learning and machine vision-based quality assurance systems across the manufacturing and packaging industries, several significant gaps remain in both academic research and industrial implementation. These gaps hinder the widespread adoption and effectiveness of intelligent inspection systems, especially in real-time, mission critical environments such as pharmaceutical, food, and consumer goods packaging.

i. **Lack of Explainable and Interpretable Models**

Most existing ML-based QA systems utilize complex, high-performance models like deep Convolutional Neural Networks (CNNs), which often operate as "black boxes" with limited transparency. Although these models achieve high accuracy in defect detection, they offer little to no insight into the reasoning behind their decisions. This poses several problems:

- **Regulatory barriers**: In safety-critical sectors such as pharmaceuticals or food, decisions must be explainable for audit and compliance purposes.
- **Operator trust**: Quality control operators and engineers are less likely to rely on automated systems if the decision-making process is not transparent.
- **Debugging and improvement**: Without understanding why a defect was detected (or missed), it's hard to refine the system or improve it over time.

ii. **Narrow Focus on Surface-Level Defects**

The majority of existing works concentrate solely on detecting superficial or surface-level defects, such as scratches, holes, or labeling errors. While these are essential, faults due to design inconsistencies, such as poor alignment, incorrect container geometry, or improper sealing design, are often ignored. These design faults may not manifest as obvious visual defects but can compromise long-term usability, product protection, or cause failures during transport and storage.

iii. **Lack of Integrated Real-Time Actuation and Sorting**

Another gap in the reviewed systems is the absence of real-time physical intervention mechanisms. Most ML-based inspection systems stop at detection they classify an item as defective but do not connect to actuators for immediate sorting, rejection, or signaling. This means that human operators still have to intervene, which reduces the efficiency and automation potential of the solution.

iv. **Insufficient Dataset Variety and Generalization:**

Several studies rely on proprietary or domain-specific datasets that do not reflect the diversity, noise, and unpredictability of real-world manufacturing environments. These limitations lead to:

- Overfitting on controlled data.
- Poor generalization to new defect types or lighting conditions.
- Difficulty in scaling or transferring the solution across different production lines or product categories.

2.4 Detailed Review of Methodology

The methodology discussed in the reviewed papers typically involves the following stages:

2.4.1 Data Acquisition and Preprocessing

Images are acquired using high-resolution cameras (e.g., in pharmaceutical bottling lines or additive manufacturing powder beds). Preprocessing includes normalization, resizing, noise reduction, and, in some cases, feature extraction using autoencoders.

2.4.2 Model Selection and Training

Supervised learning dominates the field, particularly:

- **Classification Models**: Used models like Roboflow, YOLOV8 etc. for detecting defective vs. non-defective units.
- **Segmentation and Object Detection**: Used to localize defects or regions of interest.
- **Regression Models**: For predicting product characteristics.
- **Anomaly Detection Models**: For identifying rare failure patterns.

2.4.5 Applications in Industry

- **Pharmaceuticals**: Defect detection in bottle labeling, cap sealing.
- **Additive Manufacturing**: Monitoring powder bed defects and porosity.
- **Electronics/Textiles**: Inspection of surface quality and component wear.

2.5 Summary

This chapter has provided an overview of current research on AI and XAI applications in visual quality assurance across multiple manufacturing domains. While significant progress has been made in defect detection using deep learning, broader VQA practices such as root cause analysis and predictive maintenance remain less explored. A recurring theme across studies is the need for transparent and explainable AI models to increase trust, especially in highly regulated sectors. The reviewed methodologies underscore the potential of integrating machine vision, CNNs, and XAI tools to create intelligent, self-validating inspection systems.

Chapter 3

Design & Implementation

3.1 Introduction

The implementation involved combining computer vision, deep learning, and hardware control to create a real-time defect detection and sorting system. A custom dataset of medicine packets was collected and annotated to train a YOLOv11 model using PyTorch. The model was integrated with a Python application using OpenCV for video capture and frame analysis. Each frame was processed to detect printing defects. When a faulty packet crossed a predefined line, the system sent a signal to an Arduino Uno via the PyFirmata2 library, triggering a servo motor to sort the defective item. Threading ensured that the servo operated without interrupting the video stream, enabling fast and reliable performance suitable for industrial use.

3.2 Problem Statement

In modern manufacturing, especially in developing countries like Bangladesh, maintaining persistent product quality remains a significant challenge. Orthodox manual inspection methods are increasingly incompetent for detecting packaging defects in fast-paced production lines, particularly in critical sectors like pharmaceuticals. This highlights the urgent need for a reliable, automated solution tailored to local industrial conditions.

- In developing countries like Bangladesh, quality control processes particularly defect detection and sorting are often conducted manually, relying on human vision, which is inefficient and prone to error, especially in high-speed manufacturing environments such as the pharmaceutical sector.
- Printing defects on pharmaceutical packaging (e.g, misalignments, faded text, or incorrect designs) can result in serious regulatory non-compliance and loss of consumer trust if not identified accurately and promptly.
- From a mechatronics engineering perspective, the core challenge involves integrating computer vision, machine learning, real-time decision-making, and actuator control into a reliable, automated system that performs accurately under production constraints.
- This project aims to address the issue by developing a low-cost, intelligent defect detection and sorting system that is scalable and tailored to meet the resource limitations and specific needs of local manufacturing industries.

3.3 Design Requirements & Requisites

To implement the vision-based defect detection and sorting system, several hardware, software, and functional requirements are needed. These requirements ensured that the system would be practical, cost-effective, and adaptable for industrial use, particularly in resource-limited settings like those found in many Bangladeshi factories.

3.3.1 Hardware Requirements

The successful implementation of the Vision-Based Printing Defect Detection and Sorting System required a combination of hardware components to enable real-time image analysis, decision-making, and mechanical sorting. Below is a list of the essential hardware used to build and operate the system effectively.

i. Camera (Webcam or IP Camera): To capture real-time video frames of the product packets for analysis.
ii. Arduino Nano: Used to control the servo motor based on the detection results.
iii. Servo Motor (e.g., SG90): Responsible for physically sorting defective packets when triggered.
iv. Computer/Laptop: Acts as the processing unit for running the detection algorithm and interfacing with Arduino.
v. Power Supply and Connection Wires: To ensure uninterrupted operation of all hardware components.
vi. Conveyer: To bear the medicine packets.
vii. Gear motor: To run the conveyer belt.
viii. Speed Controller: To control the speed of conveyer.

3.3.2 Software Requirements

To develop the intelligent defect detection and sorting system, several software tools and libraries were utilized to enable image processing, machine learning, and hardware communication. The following software components were essential for implementing and running the project efficiently.

i. Python 3.x: Programming language used for developing the application.
ii. PyTorch: Framework used for training and deploying the YOLOv8 object detection model.
iii. OpenCV: Library used for video capture, image processing, and frame annotation.
iv. PyFirmata2: Python library for communication between the computer and Arduino board.
v. YOLOv11 Custom Model (best.pt): A pre-trained model tailored for detecting printing defects.

3.3.3 Functional Requirements

The functional requirements of the system define the core capabilities needed to ensure accurate, real-time detection and sorting of packaging defects in a resource- efficient manner. These requirements guide the system's performance, responsiveness, and integration within a live production environment.

i. The system must detect packaging defects (e.g., 'Miss_Print') with high accuracy and confidence.
ii. Detection must be processed in real time to support continuous production lines.
iii. The servo motor must respond immediately to defective detections by initiating the sorting mechanism.
iv. The detection area must include a visual reference line to identify packets at the decision point.
v. The system should run efficiently on CPU without requiring high-end GPU resources.

3.4 Design Criteria

The design criteria involves the key performance and operational standards that the system must meet to be effective, reliable, and suitable for industrial deployment. These criteria ensure the solution not only performs accurately and efficiently but also remains adaptable, robust, and safe for continuous use in real-world manufacturing environments.

i. Accuracy: The system must detect misprinted packets with high accuracy using a confidence threshold.
ii. Real-time Performance: Detection and sorting must occur within milliseconds to support production line speed.
iii. Robustness: The system must operate reliably across different lighting conditions and packet orientations.
iv. Scalability: The system must be easily deployable in industrial environments with minimal modifications.
v. Safety: Servo movement should be smooth, controlled, and safe for integration with sorting hardware.

3.5 Design Steps

The system was developed through the following sequential steps:

a) **Dataset Collection and Annotation:**

Images of medicine packets (both normal and defective) were collected and labeled using annotation tools like RoboFlow to train a YOLOv11 model.

b) Model Training:

A custom YOLOv11 model was trained using PyTorch, focusing on identifying the 'Miss_Print' class with high precision.

c) **System Integration:**

i. The YOLO model was loaded and run on CPU to ensure broader compatibility.
ii. OpenCV was used to capture video frames and draw bounding boxes and labels.
iii. A trigger line (ROI) was defined within the video feed for decision- making.

d) Servo Control:

i. PyFirmata2 library was used to interface with the Arduino Uno.
ii. A servo motor was controlled to "unlock" and "lock" positions when a defect was detected within the decision line area.

e) Threaded Execution:

Threading was used to trigger servo movements without blocking the frame processing loop, ensuring real-time performance.

f) Testing and Validation:

The system was tested using live video feeds and recorded videos to validate accuracy, responsiveness, and servo behavior.

3.6 Design Assessment & Performance Indices

The system was assessed based on the following performance metrics and practical outcomes:

i. **Detection Accuracy:**
   The YOLO model achieved high precision in detecting misprints with minimal false positives when the confidence threshold was set to 0.5.
ii. **Processing Speed:**
   The system was able to process approximately 15–25 frames per second (FPS) on a standard CPU, meeting the requirement for real-time operation.
iii. **Servo Responsiveness:**
   Servo movements were executed within 1–2 seconds, efficiently sorting the defective packets without interrupting detection.
iv. **Reliability:**
   Over multiple test runs, the system showed consistent performance in identifying and sorting packets under various lighting conditions.
v. **Cost-effectiveness:**
   The overall setup used low-cost components (webcam, Arduino, servo), making it feasible for mass deployment in small to mid-scale industries.

3.7 Summary

The design and implementation of the project had been focused on developing a real- time defect detection and sorting system by combining computer vision, deep learning, and mechatronic control. A custom YOLOv11 model had been trained to detect misprinted labels on medicine packets. This model had been integrated with OpenCV to process live video feeds, where defects were identified based on defined confidence thresholds. To execute the sorting mechanism, PyFirmata2 had been used to establish communication with an Arduino Nano, which had been responsible for controlling a servo motor. A predefined decision line on the video frame had been used to trigger the servo whenever a defect had been detected. Threading had been applied to ensure that servo actions did not interfere with the continuous frame processing, thus enabling real-time operation. All components had been carefully selected to be low-cost and easily accessible, making the system both practical and scalable for implementation in small to medium-scale manufacturing setups in countries such as Bangladesh.

Chapter 4

Project Management & Finance

4.1 Introduction

The successful execution of any engineering project requires careful planning, time management, and resource allocation. Our project, *"Vision-Based Printing Defect Detection and Sorting System,"* was officially initiated in the third week of December, aligning with the academic calendar which provided us with 14 working weeks to complete all stages of development. Given the interdisciplinary nature of the project combining machine learning, computer vision, and hardware control effective coordination among team members was crucial. From selecting appropriate equipment to setting realistic milestones, each step had to be thoughtfully managed to ensure timely progress without compromising on quality. This section outlines how the project was organized, how time and tasks were distributed, how the necessary components were chosen, and how the overall financial plan was developed to maintain both efficiency and affordability throughout the project lifecycle.

4.2 Management of Project

Time Frame

Phase	**Duration**	**Activities Involved**
Week 1–2	Planning & Literature Review	Topic selection, requirement analysis, design planning
Week 3–5	Dataset Collection & Model Training	Image labeling, YOLOv8 model training and tuning
Week 6–8	System Integration	Connecting video feed, model inference, and Arduino setup
Week 9–11	Testing & Optimization	Real-time testing, servo timing adjustment, bug fixing
Week 12–13	Performance Evaluation	Running evaluations, collecting metrics and plotting graphs
Week 14	Documentation	Final report writing, formatting, and presentation preparation

Table 4. 1 Time Frame

Equipment Selection

Component	Purpose
Webcam (USB)	Capturing real-time video of medicine packets
Arduino Nano	Microcontroller for servo motor control
Tower Pro SG90 Pro	Actuator for physical sorting of defective packets
USB Cable & Jumpers	Connectivity between computer and Arduino
Laptop	Running Python scripts and model inference
Gear motor	To run the conveyer
Speed Controller	To control motor speed

Table 4. 2 Equipment List

Major Components

Laptop

To run the python scripts and model inference laptop was used with specification of:

- Processor Model: Ryzen 5- 5500U
- CPU Cache: 11MB
- RAM: 8GB
- RAM Type: DDR4
- Graphics Chipset: Nvidia GTX 1650 Graphics
- Graphics Memory: 4GB

Arduino Nano

The Arduino Nano is a compact, breadboard-friendly microcontroller board based on the ATmega328P. It is designed for projects that require a small form factor, low power consumption, and USB programmability. Despite its small size, it offers nearly the same functionality as the larger Arduino Uno.

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage (Recommended): 7–12V
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 8
- DC Current per I/O Pin: 40 mA
- Flash Memory: 32 KB (ATmega328P) of which 2 KB used by bootloader
- SRAM: 2 KB
- EEPROM: 1 KB

- Clock Speed: 16 MHz
- USB to Serial Chip: CH340 or FT232 (depending on version)
- Connectivity: Mini-USB port (for power and programming)
- Dimensions: 45 mm x 18 mm
- Weight: Approx. 7 grams

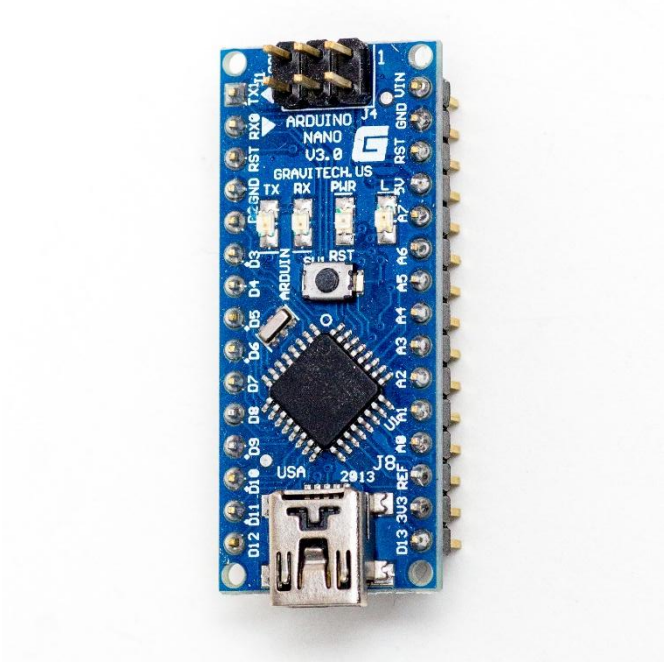


Fig 4.1: Arduino Nano

Tower Pro SG90

The **Tower Pro SG90** is a lightweight, compact analog micro servo widely used in robotics, automation projects, and hobby electronics. Despite its small size, it delivers reliable precision control and sufficient torque for tasks like sorting, positioning, and small movements.

- Operating Voltage: 4.8 V (typically powered with 5 V)
- Rotation Range: Approximately 0–180°
- Stall Torque: ~1.8 kg·cm at 4.8 V
- Operating Speed: ~0.10–0.12 seconds per 60° at 4.8 V
- Weight: Approx. 9 g
- Dimensions: ~22 mm × 12 mm × 29 mm (L×W×H)
- Deadband Width: ~7–10 µs
- Gear Type: Plastic gears
- Temperature Range: 0 °C – 55 °C
- Includes: 3 servo horns (arms) and mounting screws



Fig 4.2: Tower Pro SG90 servo motor

Webcam

For giving the model vision Fintech Luminous C30 USB 2K Quad HD 4MP Webcam was used.

- Video Resolution: 2560x 1440
- Field of view: 106°
- Frame rate: 2K/25 fps



Fig 4.3: Webcam

Gear Motor

The 25GB Gear Motor (25GA-370 model) is a compact, high-torque DC motor with a metal gearbox, designed for robotic applications such as conveyor systems, robot cars, and automation projects. It offers reliable performance with various speed options and supports both forward and reverse rotation.

- Model: 25GA-370 (25GB series)
- Operating Voltage: 12 V DC
- Available Speeds: 200 RPM / 300 RPM / 500 RPM (variant dependent)
- Rated Power: ~3.5 W
- Continuous Current: ~60 mA
- Stall Torque: Ranges from ~0.2 kg·cm to ~5 kg·cm depending on gear ratio
- Output Shaft: 4 mm diameter, ~12 mm length, D-shaped
- Motor Dimensions: Approx. 50 mm × 25 mm (body), ~70 mm including shaft
- Weight: Approx. 90–92 g
- Gear Type: Metal spur gears (durable, wear-resistant, low noise)
- Rotation Direction: Clockwise (CW) and Counterclockwise (CCW) based on polarity



Figure 4.4: Gear Motor.

Speed Controller

The PWM 1203B is a compact and efficient speed controller module designed to regulate the speed of DC motors using pulse-width modulation (PWM). It is ideal for robotics, DIY automation, and low-power motor control applications.

- Model: 1203B PWM DC Motor Speed Controller
- Input Voltage Range: 6 V to 28 V DC
- Maximum Output Current: Up to 3 A
- Maximum Power: 80 W (at 28 V, 3 A)
- PWM Duty Cycle Range: Adjustable from approximately 5% to 100%
- Control Interface: Manual potentiometer knob
- Connection: Screw terminals for motor and power connections
- Indicator: Power-on LED
- Dimensions: Approx. 50 mm × 32 mm × 15 mm
- Weight: Around 18–25 g
- Direction Control: Motor direction can be changed by reversing polarity

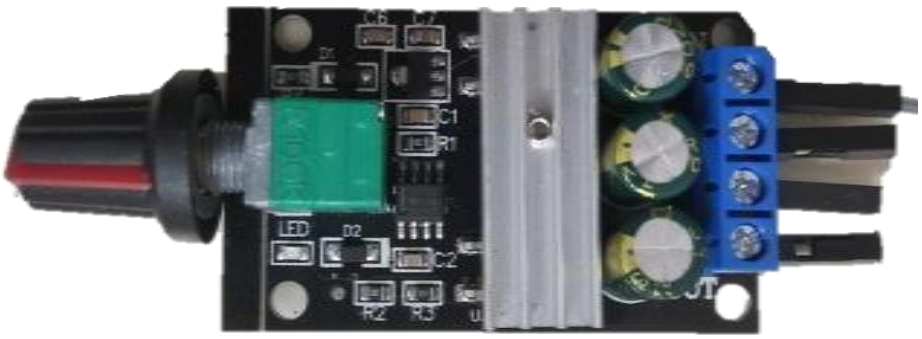


Figure 4.5: Speed Controller.

Overall Budget

Item	Price (BDT)
Dataset	4000
Arduino Nano (1)	700
Tower Pro SG90 Pro (1)	500
Webcam (HD USB) (1)	3500
Cables, Breadboard, etc.	600
Miscellaneous (printing, stationery)	1500
Gear Motor (1)	450
Speed Controller (1)	660
Bearings (4)	400
PVC Sheets	450
PVC Board	250
PVC Pipe	130
Motor Shaft	100
Additives (super glue,M seal)	400
Emery Paper	200
Wirings	360
Power Suppliers	450
Total Cost	14,650

Table 4. 3 Overall Budget

Purchasing Procedure

To procure the components required for the Vision-Based Printing Defect Detection and Sorting System, the purchasing process began with identifying reliable local and online vendors for electronic and mechanical parts. The Arduino Nano, Tower Pro SG90 servo motor, HD USB webcam, gear motor, and speed controller were sourced from trusted electronics marketplaces, ensuring compatibility and availability of warranties. Supporting items such as cables, breadboard, wiring, power supplies, and miscellaneous materials (like printing and stationery) were purchased from local electronics stores. For the mechanical setup, components like bearings, PVC sheets, PVC board, PVC pipes, motor shaft, and additives (such as super glue and M-seal) were collected from hardware stores specializing in DIY and robotics materials. Additional tools like emery paper were included for finishing and fitting adjustments. All purchases were made based on project specifications, quality, cost-effectiveness, and availability, and the total cost was calculated after collecting individual receipts to ensure proper budgeting and financial transparency.

4.3 Summary

The project was undertaken on an organized schedule of 14 working weeks with the third week of December as the start and the academic calendar adjusted accordingly. The most important steps, including planning, dataset preparation, system integration, testing, and documentation, were planned and implemented effectively.The most important hardware (e.g. the Arduino Uno, servo motor and webcam) was chosen according to their cost, availability and compatibility to open source software frameworks. The overall estimated project cost was affordable in the range of.Ensuring financial viability of 14,500 to 15000, and organizing time, careful selection of resources and a budget-efficient cost, the project was successfully implemented, which resulted in the deployment of a functional, real-time defect-detecting and sorting system.

Chapter 5

Addressing Complex Engineering Problem & Activities

5.1 Introduction

The educational field of engineering, especially mechanical-informatics, was also not limited to theoretical studies, but students were supposed to be capable of solving complex, real-world challenges by means of interdisciplinary knowledge and practice. In this project, it was found out that the on-textbook knowledge of such issues demanded critical thinking, technical integration, and innovative solution based on hands on experience. This project was done as a representative of this learning objective, Vision Based Printing Defect Detection and Sorting System. The engineering principles that were necessary were numerous, and the contributions that were made in the project through artificial intelligence, real time computing, computer vision, embedded systems and electromechanical control all had to be put together to make a coherent and workable solution [1]. All the work that was done on this project was done with planning, trial and error and constant refinement. A variety of complex engineering issues were met and resolved in a systematic manner and technical work was done according to the real-life industrial requirements. This chapter provides a review of how the solution to those problems was to be done, and how the principles of engineering were practically applied in order to have a working and efficient system.

5.2 Addressing Complex Engineering Problem

5.2.1 Teamwork

This project could not have been successful without effective coordination of team members who had different skills. The tasks were shared according to each person by his/her strengths which included preparing datasets, training models, integrating hardware and testing. Meetings frequently took place in order to get the goals in line and to troubleshoot together and bring every aspect of the system into line. The multidisciplinary character of the project (mechanical, electrical, and software engineering) demanded excellent coordination and support to each other to resolve the design and integration issues.

5.2.2 Communication

Both the project lifecycle and communication were well-maintained. The team members tracked the tasks and kept in touch with any progress via digital tools (e.g., WhatsApp, Google Docs, Trello). Technical documentation, the code comments, and version control made sure that all people were aware of the logic and the alterations in the system. The results and issues during development were formed in both technical and non-technical terms to make the results easy to understand in various fields.

5.2.3 Project Management and Finance

Simple project management rules were adhered to to structure the work, establish timeframes, and deal with the risks. To break down the project into small manageable components, a work breakdown structure (WBS) was established that included dataset preparation, training, testing, setting up hardware and integration. There was a timeline to watch progress. Financially, a tight budget was adhered to when acquiring Stuff like the Arduino Nano, servo motor, gear motor and webcam. The comparison of local suppliers was done to be cost effective and all the purchases were monitored and documented to remain within the budget allocated.

5.2.4 Life Long Learning

The team members in the project were involved in lifelong learning in order to acquire new technologies and methodologies. Online courses, documentation, and tutorials were read to know about YOLOv11 nano, training PyTorch models, serial communication, and Arduino programming. This education was not restricted to the classroom and it promoted use of modern tools and structures. The issues of processing real time video and motor control prompted the team to consider new problem-solving strategies and keep abreast with computer vision and embedded system development.

5.3 Summary Table

Work Profile	Description
1. Engineering Knowledge	Applied interdisciplinary knowledge including electronics (Arduino, motor control), computer science (Python, machine learning), and mechanical systems (conveyor, servo) to build an intelligent detection and sorting mechanism.
2. Problem Analysis	Analyzed inefficiencies in manual inspection processes. Broke the problem into image classification, real-time detection, and mechanical sorting. Considered constraints like accuracy, processing speed, and cost.
3. Design	Designed a full working pipeline involving custom dataset creation, model training, conveyor setup, camera placement, real-time inference, and servo based actuation for dynamic sorting of medicine packets.
4. Investigation	Researched various models and tools to identify YOLOv11 nano for lightweight detection. Experimented with hardware configurations and environmental conditions to ensure consistent results across scenarios.
5. Modern Tool Usage	Used Python, PyTorch, OpenCV, YOLOv11, Arduino IDE, and PyFirmata2 to develop and connect all parts of the system. These tools enabled real-time video inference and precise motor control via serial communication.
6. Engineers and Society	Addressed a critical issue in the pharmaceutical industry by preventing mislabeled or defective packets from reaching consumers. Improved product safety, reduced human error, and promoted industrial automation in local contexts.
7. Environment & Sustainability	Developed a low cost, energy efficient solution with minimal hardware that supports reuse and scalability. Reduced industrial waste and human labor dependency, promoting eco-friendly manufacturing practices.
8. Ethics	Ensured accurate and safe classification of packets, particularly in a health sensitive industry. Maintained data integrity during training and upheld transparency and responsibility in system performance and deployment.

5.4 Addressing Complex Engineering Activities

5.4.1 Teamwork

The success of this project depended heavily on effective collaboration among team members with diverse skill sets. Responsibilities were distributed based on individual strengths such as dataset preparation, model training, hardware integration, and testing. Regular meetings were held to align goals, solve problems collaboratively, and ensure all parts of the system were synchronized. The multidisciplinary nature of the project (combining mechanical, electrical, and software engineering) required strong coordination and mutual support to overcome design and integration challenges.

5.4.2 Communication

Clear and consistent communication was maintained throughout the project lifecycle. Team members used digital platforms (e.g., WhatsApp, Google Docs, Trello) for task tracking and progress updates. Technical documentation, code comments, and version control ensured that everyone understood the logic and changes in the system. During development, results and issues were presented in both technical and non-technical terms to facilitate understanding across different domains. Communication was also crucial for preparing reports and presenting outcomes to mentors or evaluators.

5.4.3 Project Management and Finance

Basic project management principles were followed to organize tasks, set deadlines, and manage risks. A work breakdown structure (WBS) was created to divide the project into smaller, manageable units such as dataset preparation, training, testing, hardware setup, and integration. A timeline was maintained to monitor progress. From a financial perspective, a strict budget was maintained for purchasing components such as the Arduino Nano, servo motor, gear motor, and webcam. Local suppliers were compared to ensure cost effectiveness, and all purchases were tracked and recorded to stay within the allocated budget.

5.4.3 Lifelong Learning

Throughout the project, team members engaged in continuous learning to understand new technologies and methodologies. Tutorials, online courses, and documentation were consulted to learn about YOLOv11 nano, PyTorch model training, serial communication, and Arduino programming. This learning extended beyond classroom knowledge, encouraging the adoption of modern tools and frameworks. Challenges such as real-time video processing and motor control motivated the team to explore new problem solving approaches and stay updated with advancements in computer vision and embedded systems.

5.5 Summary Table

Complex Engineering Activity	Description
1. Teamwork	Worked in a multidisciplinary team sharing duties, including database preparation, model training, hardware integration, and testing. The team members assisted one another in the development cycle to have a smooth strategy in coordination and project completion.
2. Communication	Ensured good communication through technological means such as the WhatsApp, Google Docs etc. Frequent updates and record keeping were also useful in making sure that the entire team was in line with project objectives and technical activities.
3. Project Management and Finance	Practical underlying project management concepts like work breakdown structure (WBS) development, milestones and expense tracking. The purchase of components was done within a defined budget and developments were tracked in order to deliver on time.
4. Lifelong Learning	Constantly acquired and used new skills, such as how to train YOLOv11 models, how to interface Arduino with Python, and how to process video with OpenCV. The online tutorials and documentation were referred to regularly to address technical challenges to keep up with the changing tools.

5.6 Summary

The engineering problem that was tackled in the project Vision-Based Printing Defect Detection and Sorting System involved a combination of knowledge in various field of engineering software, electronics as well as mechanical systems to achieve an intelligent real time defect printing detection system in pharmaceutical packaging. The task was challenging because it involved real-time processing, dynamic classification, hardware-software interfacing and low-resource operation. A solution to this would be to train a custom YOLOv11 model on a manually curated dataset of medicine packets images, and combine the OpenCV with it to process live video. The system then connected to an Arduino Nano through serial communications to activate a servo motor to sort defective packets in a conveyor belt. The whole arrangement was designed to be efficiently run on a typical laptop processor and it was scalable and affordable to local industry. To overcome this challenge, a number of sophisticated engineering efforts were undertaken. Close collaboration was portrayed by distribution of roles and shared problem solving. There was effective and consistent communication through the use of digital tools and documentation practices. Project management and financial planning were adopted to provide on schedule progress and cost-effective component acquisition. The team also undertook lifelong learning, constantly learning new skills in deep learning, embedded systems, and real-time automation.

Chapter 6

Result and Analysis

6.1 Introduction

The aim of the chapter was to compare the effectiveness and reliability of the designed vision-based system named Vision-Based Printing Defect Detection and Sorting System by assessing its functionality in terms of conventional machine learning evaluation methods. The techniques involved a visual evaluation of prediction outputs and a constant evaluation on the basis of different performance measures. The confusion matrix, which showed the accuracy of classification across the classes (Good, Miss_Print, and Background); the precision-recall (PR) curve, which measured the trade off between the correct positive prediction and coverage of actual positives; the F1-confidence curve, which indicated the optimal trade off between precision and recall at different levels of confidence; and the recall-confidence curve which monitored the capability of the model to identify defects as the confidence level varied was also determined to be of utility in this analysis. These analyses combined gave a comprehensive opinion on the strengths, limitations, and preparedness of the model to be used in reality in industries.

6.2 Results & Analysis

i. Confusion Matrix

- The normalized confusion matrix shows the model's ability to distinguish between the classes: Good**,** Miss_Print**,** and Background**.**
- True Positive Rate for 'Good' class**:** 94%
- True Positive Rate for 'Miss_Print' class**:** 77%
- The classification errors mainly occurred due to overlapping features or noisy backgrounds.

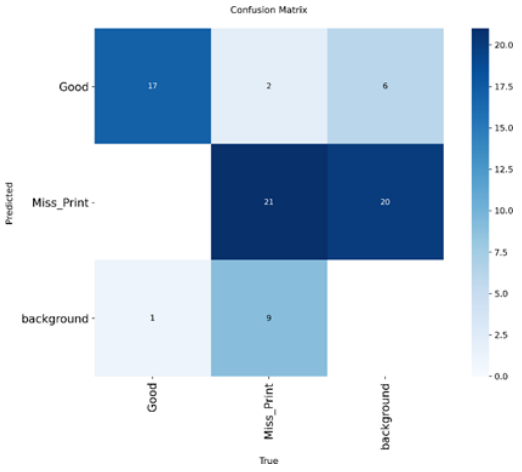


Fig6.1: Confusion Matrix

ii. Precision-Recall (PR) Curve

- Precision for Good: 0.898
- Precision for Miss_Print: 0.649
- mAP@0.5 (All classes): 0.774
- The model achieved high precision in identifying good products, although precision for defects (Miss_Print) was slightly lower due to dataset variability.

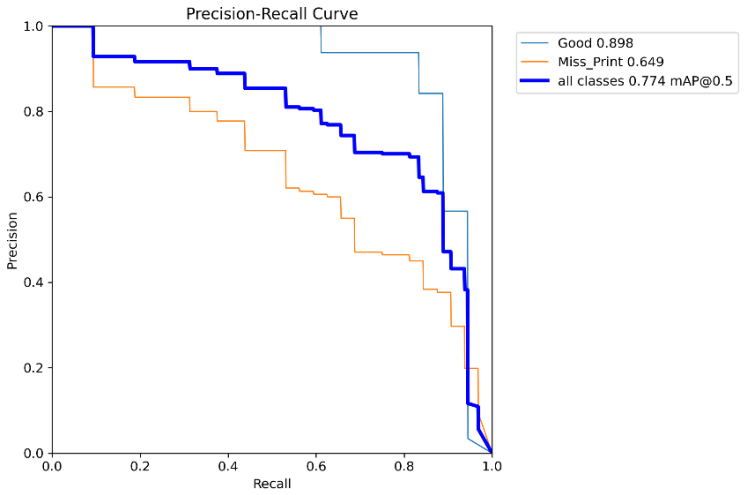


Fig6.2: PR Curve

iii. F1-Confidence Curve

- Best F1 score (overall)**:** 0.72 at confidence 0.389
- F1 scores for 'Good' were higher across confidence thresholds, indicating better overall performance in detecting defect-free packets.
- The curve reflects a trade-off between recall and precision, with optimal confidence around 0.38–0.4.

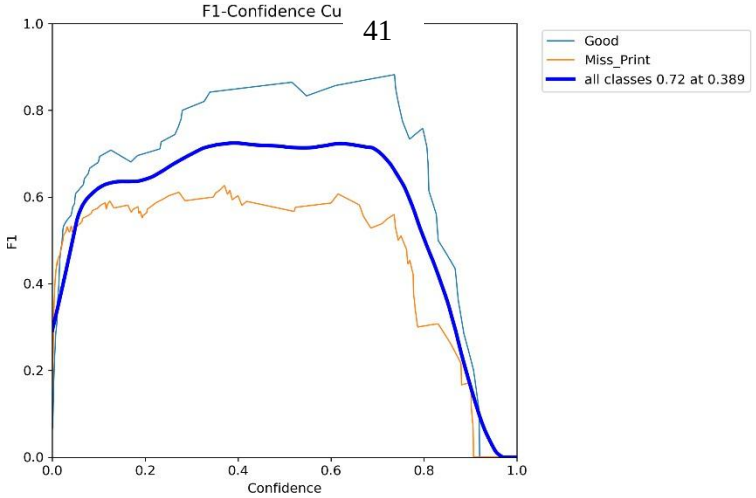


Fig 6.3: F1-Confidence Curve

iv. **Recall-Confidence Curve**

- Overall recall: ~94% at confidence near 0.0
- The recall for ‘Good’ class remains consistently high, while ‘Miss_Print’ has moderate recall values (~60–70%), suggesting some misprints were less confidently detected.

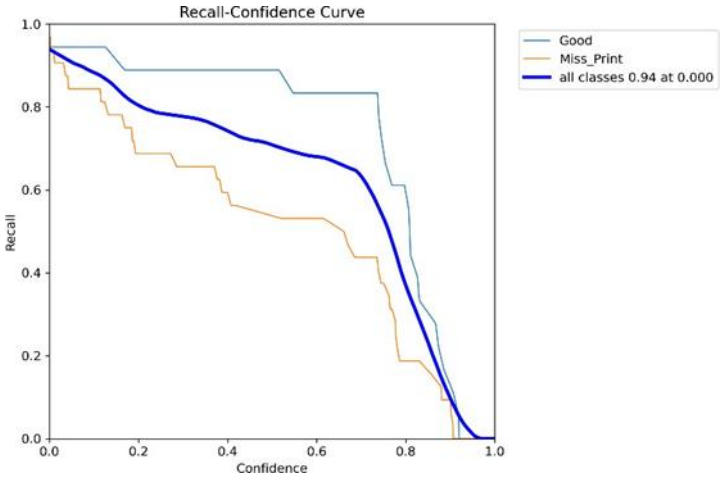


Fig 6.4: Recall-Confidence Curve

6.1 Summary

The result and analysis chapter demonstrated that the developed system performed reliably in detecting printing defects on packaging. The confusion matrix revealed reasonable classification capability, with a normalized accuracy of 94% for correctly identifying Good packets and around 77% for Miss_Print packets. The precision- recall curve showed a strong performance with an average precision of 89.8% for Good and 64.9% for Miss_Print, indicating a higher reliability in detecting properly printed items. The F1-confidence and recall-confidence curves further validated the model’s stability across varying confidence thresholds, with the overall F1 score peaking at 0.72 and recall at 0.94. These metrics confirmed that the system maintained a good balance between detecting faults and minimizing false alarms, making it suitable for real-time deployment in industrial settings.

Chapter 7

Socio-economic Impact on Sustainability

7.1 Introduction

Sustainability in the current industrial environment is no more an environmental issue but a multi dimensional concept that addresses economic efficiency, social responsibility, health, safety, and ethical practices that are legal [10]. With the emergence of mechatronics engineering projects looking to adopt more sophisticated technologies including machine learning and automation, it is necessary to assess their overall socio economic effects. The project was designed under the title Vision-Based Printing Defect Detection and Sorting System not to only solve an engineering problem, but also to make significant contribution to the sustainable development of industrial automation and the society, in particular, in the developing nations such as Bangladesh where resource optimization and labor welfare is a major issue.

7.2 Impact on Societal, Health, Safety, Legal, and Cultural Issues

7.2.1. Societal Impact

Introduction of automated defect detection systems has the potential of radically changing the packaging and quality control industries. Conventionally, these are carried out using human input that is labor intensive, prone to errors and is usually linked to worker fatigue and low productivity. This can be automated and thus the laborers can be relieved leading to the human resources being reallocated in the supervisory or skilled department. The shift is a social upliftment factor because of skill development and lessening repetitive strain injury (RSIs) particularly in factory workers.
[11].

7.2.2. Health and Safety

Manual inspection of printing defects, particularly in pharmaceutical or electronic industries, often exposes workers to long hours, stress, and sometimes harmful environments like ink fumes or dust. The adoption of this automated system reduces direct human exposure to such occupational hazards. Moreover, the improved accuracy in defect detection ensures that faulty or mislabeled medicine packages are not circulated, directly safeguarding public health [12].

7.2.3. Legal and Regulatory Compliance

In industries like pharmaceuticals and food processing, packaging quality is strictly regulated by national and international standards (e.g., ISO 9001, GMP). Incorrect labeling or print errors may result in regulatory non-compliance, leading to legal consequences or product recalls. By integrating machine learning-based visual inspection, companies can ensure compliance with quality assurance protocols and avoid such costly legal implications (Bangladesh Standards and Testing Institution – BSTI, 2022) [13].

7.2.4. Cultural Acceptance and Workforce Adaptation

The transition to automation in a country like Bangladesh must consider cultural and workforce sensitivities. While automation may raise concerns about job losses, the project's implementation model promotes "collaborative automation", where machines assist rather than replace human workers. With proper training and awareness programs, the workforce can be empowered to operate and maintain such systems, encouraging cultural acceptance of technology-enhanced workplaces [14].

7.2.5. Economic Efficiency

Economically, this project contributes to sustainability by reducing inspection time, minimizing waste from defective packaging, and lowering labor costs. The improved consistency in quality also enhances brand reputation and consumer trust, resulting in long-term economic gain. For small and medium industries in Bangladesh, such affordable AI-integrated solutions could offer a competitive edge in the global market while supporting the UN Sustainable Development Goals (SDG 8: Decent Work and Economic Growth, SDG 9: Industry, Innovation, and Infrastructure).

7.3 Sustainability Impact

7.3.1 Resource Efficiency and Waste Reduction

Among the most direct project sustainability contributions, there is the capability to reduce material waste. In conventional manufacturing settings, particularly in the packaging sector, defects are usually detected late at night- a batch of products is thrown away, and a lot of materials go to waste in the form of paper, ink, plastic and labor. This system can be used to identify defective packets early on with the help of computer vision and real-time defect detection, and such early detection will enable removing the packets in time and minimize the necessity of reprocessed materials. This does not only save on raw materials, but also simplifies production cycles, thus manufacturing is less resource consuming and environmentally demanding.

7.3.2. Low Energy Consumption and Hardware Sustainability

The system was designed deliberately with low-power parts including: Arduino Nano, SG90 motorized servo, and a simple HD USB web camera. These parts use little power as opposed to the expensive industrial automation systems, so the solution can be used in factories with small electrical infrastructure. Furthermore, the project can execute successfully on a CPU based laptop without the use of expensive and power-intense GPUs. This is sustainable in energy and is effective to be applied in rural or semi-urban places where power supply can be unpredictable or scarce.

7.3.3. Open-Source and Cost-Effective Design

Sustainability does not focus on energy and materials only, but also on economic and technological availability. The platform used in this project includes the open-source Python, PyTorch, OpenCV, and PyFirmata2 that do not need the expensive subscription or licence. Consequently, the whole system can be upgraded, modified and serviced by local engineers or students, which guarantees long-term use and experience.Transfer without dependence on external proprietary solutions. Low manufacturing cost of components also enables mass copying in small and medium enterprises (SMEs), which help in larger scale modernization of industries.

7.3.4. Environmental Impact and Industrial Sustainability

The project indirectly minimizes carbon emission through the remanufacture and disposal of waste packaging since the number of defective products decreases. In addition, the number of erroneous packets decreases, minimizing the transfer of unproductive goods, thereby decreasing fuel use and logistics emissions. This simplified process leads to cleaner production cycle and assists companies to reduce their environmental footprint without breaking international environmental standards and laws.

7.3.5. Alignment with UN Sustainable Development Goals (SDGs)

This project directly leads to SDG 9 Industry, Innovation, and Infrastructure because it will encourage sustainable industrialization through quality control and automation innovation. It is also in line with SDG 12 Responsible Consumption and Production because it minimizes waste, enhances efficient use of resources and promotes environmentally friendly production behavior. The system promotes the sustainability of economic growth by injecting intelligent and scalable solutions to the local industries without sacrificing the future.

7.3 Summary

The Vision-Based Printing Defect Detection and Sorting System has shown great socio-economic importance through quality enhancement of products, lessening of dependence on manual labor and increased safety during the pharmaceutical packaging procedure. It helps to upgrade the workforce, adhere to regulations, and promote population health by making sure that only properly labeled medicine packets are made available to consumers. Economically, the project increases the efficiency of production and reduces the loss caused by the defective packaging, which benefits the local industries in Bangladesh in terms of competitive advantages. In terms of sustainability, the project enhances resource efficiency and reduces energy use and material wastage caused by detecting defects at the early stages. This is due to the fact that the affordable, open-source technologies are used, which means the long-term scalability and accessibility especially to the small and medium enterprises. On the environmental front, it contributes to cleaner production and global development SDG 9 (Industry, Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production) and as a result, is a socially responsible and environmental conscious engineering solution.

Chapter 8

8.1 Conclusion

The project was able to successfully meet its main goals in designing and developing a vision system based on machine learning that allows identifying defects in printing of medicine packets with a high degree of accuracy. The system was effective in detecting misalignment, faded text and erroneous designs by utilizing a sophisticated computer vision method, which also included automated sorting of the printed materials via a real-time hardware reaction system with Arduino and servo motors. This greatly lowered the reliance on human labor, minimized the human error as well as enhanced the production line efficiency that proves the usefulness of intelligent automation in practice. Lastly, the fully operational prototype system was designed and tested and established the viability of implementing low cost scalable systems into the real world. Not only does the successful completion of this project satisfy the stated objectives, but also provides the basis of future enhancement and industrialization of the vision-based defect detection technologies.

8.2 Limitations

Although the project was able to meet its goals and also present a working prototype, during the development and testing, it was noted that there were several limitations. These shortcomings present some limits within which system performance, scalability and adaptability could be limited under certain conditions. Such challenges can only be comprehended to lead to future enhancements and to make the system even more applicable in other industrial environments.

I. Some of the test settings had moderate effects on the performance of the model due to changes in lighting and clutter on the background.
II. Real-time speed can be limited by CPU capacity; on systems without GPU acceleration, it can be reduced by performance.
III. The system was trained using a comparably small and specific-domain dataset, which constrained its application to other classes of packaging or defects.
IV. Servo-based sorting is best suited to low-speed production lines; it may need additional mechanical optimization to be integrated with high-speed conveyor systems.
V. The system is currently binary (Good vs. Miss_Print), and the other possible complex or compound defect types might escape detection.

8.3 Future Research Directions

a. **Dataset Expansion and Generalization**: Future research could involve collecting a larger, more diverse dataset across different packaging types, printing technologies, and defect categories to improve model robustness and domain transferability.

b. **Integration with Robotic Arms or Conveyor Systems**: Instead of a single servo motor, more advanced electromechanical sorting methods like robotic pick-and-place arms or conveyor diverters could explore.

c. **Edge Deployment and Model Optimization**: Converting the model to run efficiently on edge devices (e.g., NVIDIA Jetson Nano, Raspberry Pi with Coral TPU) can reduce latency and enhance portability for field deployment [3].

d. **Multiclass Defect Detection**: Future models could classify multiple defect types such as color mismatch, text misalignment, missing labels, or physical damage, improving industrial quality control.

e. **Integration with IoT and Real-Time Monitoring Dashboards**: Embedding the system within an IoT framework could enable remote monitoring, data logging, and cloud-based quality analytics, enhancing traceability and decision-making.

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